

## SVANBERGA, Sweden

WMO: 024900

Lat: 59.8317N

Lon: 18.6313E

Elev: 16

StdP: 101.13

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 2             | -20.5      | -16.1 | -22.6                         | 0.5 | -20.4 | -18.1 | 0.8 | -16.0 | 6.4                   | -3.2 | 5.8 | -3.1 | 0.2                   | 270  | 0.553 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.4                   | 26.3            | 19.0 | 24.4 | 17.9 | 22.8 | 17.0 | 20.2                | 24.1 | 19.0 | 22.8 | 18.0 | 21.5 | 2.4                  | 150  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 18.8                            | 13.7 | 21.8 | 17.6 | 12.7 | 20.8 | 16.5 | 11.8 | 19.8 | 58.1          | 24.3 | 54.1 | 22.7 | 50.8 | 21.5 | 24.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 5.7           | 5.0  | 4.3 | DB                         | -25.5 | 28.6               | 5.1 | 1.3                                                | -29.2 | 29.5       | -32.1 | 30.3       | -35.0 | 31.0       | -38.7 | 31.9 | (4) |
| (5)               |      |     | WB                         | -24.0 | 21.4               | 5.4 | 1.7                                                | -27.9 | 22.6       | -31.1 | 23.6       | -34.1 | 24.5       | -38.1 | 25.7 | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 5.9    | -3.4  | -3.7  | -0.8  | 4.1   | 9.4   | 13.2  | 16.6  | 15.4  | 11.3  | 6.2   | 2.3   | -1.0 |
|                                                                                  | DBStd       | 8.05   | 5.50  | 5.42  | 4.98  | 3.37  | 3.33  | 2.79  | 2.67  | 2.62  | 2.95  | 3.65  | 3.71  | 5.09 |
|                                                                                  | HDD10.0     | 2091   | 414   | 384   | 337   | 179   | 52    | 5     | 0     | 1     | 20    | 127   | 231   | 342  |
|                                                                                  | HDD18.3     | 4573   | 673   | 617   | 594   | 426   | 276   | 154   | 68    | 95    | 211   | 377   | 481   | 600  |
|                                                                                  | CDD10.0     | 578    | 0     | 0     | 0     | 3     | 33    | 101   | 204   | 169   | 59    | 8     | 0     | 0    |
|                                                                                  | CDD18.3     | 21     | 0     | 0     | 0     | 0     | 0     | 1     | 14    | 6     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 283    | 0     | 0     | 0     | 0     | 7     | 46    | 172   | 58    | 1     | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 41     | 0     | 0     | 0     | 0     | 1     | 6     | 28    | 6     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 1.9    | 2.1   | 1.9   | 2.0   | 2.0   | 2.0   | 1.8   | 1.8   | 1.7   | 1.8   | 1.9   | 2.1   | 2.1  |
| Precipitation                                                                    | PrecAvg     | 586    | 41    | 35    | 29    | 30    | 39    | 56    | 63    | 68    | 49    | 53    | 62    | 56   |
|                                                                                  | PrecMax     | 772    | 76    | 61    | 65    | 74    | 59    | 92    | 143   | 175   | 95    | 107   | 113   | 111  |
|                                                                                  | PrecMin     | 441    | 8     | 15    | 2     | 7     | 9     | 19    | 18    | 9     | 11    | 15    | 20    | 15   |
|                                                                                  | PrecStd     | 74     | 20    | 13    | 16    | 18    | 14    | 23    | 30    | 40    | 21    | 23    | 26    | 24   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 7.0   | 7.7   | 13.9  | 19.5  | 24.1  | 27.4  | 29.3  | 27.4  | 22.4  | 16.2  | 11.3  | 10.5 |
|                                                                                  |             | MCWB   | 5.2   | 4.8   | 8.0   | 11.3  | 14.1  | 17.5  | 19.7  | 20.5  | 16.7  | 13.1  | 10.1  | 8.3  |
|                                                                                  | 2%          | DB     | 5.5   | 5.4   | 10.1  | 16.7  | 21.3  | 24.4  | 27.2  | 24.9  | 19.9  | 14.3  | 9.9   | 8.0  |
|                                                                                  |             | MCWB   | 4.3   | 3.6   | 5.8   | 10.0  | 13.8  | 16.4  | 19.7  | 18.5  | 15.5  | 12.2  | 8.8   | 6.5  |
|                                                                                  | 5%          | DB     | 4.4   | 4.2   | 7.6   | 13.9  | 19.3  | 22.5  | 25.2  | 23.2  | 18.2  | 13.0  | 8.8   | 6.4  |
|                                                                                  |             | MCWB   | 3.4   | 2.8   | 4.4   | 8.2   | 12.7  | 15.7  | 18.9  | 17.8  | 14.6  | 11.4  | 7.8   | 5.2  |
|                                                                                  | 10%         | DB     | 3.1   | 2.9   | 5.6   | 11.4  | 17.2  | 20.8  | 23.4  | 21.7  | 16.7  | 11.8  | 7.6   | 5.1  |
|                                                                                  |             | MCWB   | 2.3   | 1.9   | 3.3   | 6.7   | 11.6  | 14.9  | 18.0  | 17.0  | 13.8  | 10.4  | 6.8   | 4.2  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 5.9   | 5.2   | 8.1   | 12.0  | 16.4  | 19.0  | 22.2  | 22.1  | 17.3  | 13.8  | 10.3  | 9.1  |
|                                                                                  |             | MCDB   | 6.7   | 6.7   | 13.3  | 18.0  | 21.9  | 24.6  | 25.9  | 25.8  | 20.5  | 15.2  | 10.7  | 10.5 |
|                                                                                  | 2%          | WB     | 4.5   | 4.0   | 6.1   | 10.2  | 14.7  | 17.6  | 20.7  | 19.9  | 16.1  | 12.5  | 8.9   | 6.7  |
|                                                                                  |             | MCDB   | 5.3   | 5.2   | 9.2   | 15.4  | 19.6  | 22.6  | 25.0  | 22.9  | 18.8  | 13.8  | 9.6   | 7.6  |
|                                                                                  | 5%          | WB     | 3.5   | 3.0   | 4.8   | 8.7   | 13.3  | 16.4  | 19.6  | 18.6  | 15.2  | 11.6  | 7.7   | 5.4  |
|                                                                                  |             | MCDB   | 4.4   | 3.9   | 7.1   | 13.3  | 18.2  | 21.2  | 23.7  | 21.6  | 17.3  | 12.8  | 8.4   | 6.1  |
|                                                                                  | 10%         | WB     | 2.4   | 1.9   | 3.6   | 7.2   | 12.1  | 15.4  | 18.6  | 17.6  | 14.2  | 10.6  | 6.8   | 4.2  |
|                                                                                  |             | MCDB   | 3.0   | 2.9   | 5.5   | 10.7  | 16.2  | 19.7  | 22.5  | 20.6  | 16.3  | 11.5  | 7.5   | 4.9  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 5.9   | 7.6   | 9.1   | 11.3  | 11.8  | 11.7  | 11.4  | 10.8  | 9.1   | 7.2   | 4.9   | 5.2  |
|                                                                                  |             | MCDBR  | 5.1   | 7.1   | 11.4  | 16.7  | 16.2  | 16.2  | 14.5  | 13.4  | 11.5  | 8.4   | 5.6   | 5.3  |
|                                                                                  |             | MCWBR  | 4.9   | 5.4   | 7.5   | 10.2  | 9.4   | 9.1   | 7.9   | 7.9   | 7.7   | 6.4   | 5.3   | 5.1  |
|                                                                                  | 5% WB       | MCDBR  | 5.1   | 6.1   | 10.2  | 14.9  | 13.9  | 14.4  | 12.4  | 11.5  | 10.4  | 7.3   | 5.7   | 5.2  |
|                                                                                  |             | MCWBR  | 4.9   | 5.0   | 7.1   | 9.4   | 8.6   | 8.5   | 7.2   | 7.3   | 7.3   | 5.9   | 5.4   | 5.1  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.280  | 0.318 | 0.337 | 0.373 | 0.371 | 0.362 | 0.389 | 0.379 | 0.347 | 0.334 | 0.304 | 0.263 |      |
|                                                                                  | taud        | 2.208  | 2.283 | 2.346 | 2.328 | 2.399 | 2.461 | 2.406 | 2.427 | 2.505 | 2.416 | 2.242 | 2.123 |      |
|                                                                                  | Ebn at Noon | 500    | 681   | 794   | 826   | 857   | 871   | 834   | 813   | 785   | 663   | 459   | 361   |      |
|                                                                                  | Edh at Noon | 42     | 68    | 86    | 105   | 105   | 100   | 104   | 95    | 75    | 60    | 41    | 30    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.26   | 0.80  | 2.31  | 4.00  | 5.39  | 6.01  | 5.60  | 4.39  | 2.76  | 1.19  | 0.35  | 0.17  |      |
|                                                                                  | RadStd      | 0.03   | 0.15  | 0.24  | 0.36  | 0.44  | 0.36  | 0.52  | 0.36  | 0.29  | 0.19  | 0.08  | 0.02  |      |

## Historical Trends

|                              |  | DBAvg | Heating |        |       | Cooling |       |      | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|-------|---------|-------|------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |      | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j)  | (k)         | (l)     | (m)     |         |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A  | N/A         | N/A     | N/A     |         |
| (47) Regional (47 neighbors) |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | -112 | -137        | N/A     | N/A     |         |

Nomenclature: See separate page